Comparative Evaluation of nnU-Net v2 and Lightweight U-Net Architectures for Liver Tumor Segmentation

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1 Project Introduction and Motivation

Tumor segmentation from computed tomography (CT) images plays a critical role in clinical diagnosis, treatment planning, radiotherapy guidance, and disease progress monitoring (Sun et al., 2019). An accurate delineation of tumor boundaries enables quantitative assessment of tumor progression and supports personalized cure strategies (Akakuru et al., 2022). However, manual segmentation by radiologists is labor-intensive, time-consuming, and an observer variant, and so motivates the development of automated segmentation methods (Nair et al., 2025).

Despite recent advances in medical image analysis, accurate and reliable tumor segmentation in CT scans remains challenging due to heterogeneity in tumor morphology, low contrast between tumor and surrounding tissues, imaging artifacts, inter-subject variability (Abdoli et al., 2026). In addition, some tumors may exhibit complex shapes and multi-scale spatial characteristics, making robust feature extractions difficult for conventional approaches (Abdusalamalov et al., 2023).

Deep learning-based segmentation frameworks, particularly convolutional neural networks (CNNs), have demonstrated strong performance in medical image segmentation tasks (Mienye et al., 2025). Architectures such as U-Net and its variants have become widely adopted due to their encoder-decoder structure and ability to capture contextual information (Yin et al., 2022) More recently, transformer-based and hybrid CNN-transformer models have further improved segmentation performance by modeling long-range spatial dependencies and multi-scale representations (Yousaf et al., 2026).

Although recent deep learning frameworks have achieved strong segmentation performance, many state-of-the-art models require substantial computational resources and long training times. In practical settings, lightweight architecture will offer advantages in terms of efficiency, interpretability, and ease of deployment while still achieving competitive segmentation performance (Rasheed, 2026). Motivated by this tradeoff, our work focuses on comparing a strong existing segmentation framework with simpler custom-designed architectures for liver tumor segmentation in CT images.

In this work, we first utilized nnU-Net to perform supervised semantic segmentation on CT scans of the liver, with the purpose of segmenting tumors. We began by following nnU-Net's preprocessing pipeline, followed by using a 3D nnU-Net with 5-fold cross validation. We analyzed its performance and compared it with that of our customized lightweight 2D segmentation architecture based on an encoder-decoder U-Net Design with skip connections. The custom model processes individual 2D axial CT slices to reduce memory usage and training time. These 2D slice segmentation were stacked to find the volume.

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2 Related Work

Deep learning-based segmentation such as CNN and U-Net based architectures have become a dominant approach for tumor delineation in CT imaging. Ronneberger et al. introduced the U-Net architecture in 2015, which had an encoder-decoder design with skip connections. The model showed strong performance in biomedical segmentation tasks even with limited annotated data through extensive data augmentation (Ronnenberger et al., 2015). This has since become the foundation for medical image segmentation tasks. More broadly, Rai notes that CNN and U-Net based methods have shown clear advantages in semantic segmentation for cancer imaging, though further research is still needed to address image quality variability across imaging systems (Rai, 2024).

The nnU-Net framework built upon U-Net by automating many of the manual tuning aspects, such as preprocressing, network configuration, training strategy, and inference settings, which enables nnU-Net to adapt itself to the properties of a given dataset. This adaptability has made it a widely used baseline for medical image segmentation tasks (Isensee et al., 2021). Our work utilizes the common supervised segmentation paradigm by applying nnU-Net to CT scans of the liver, with the purpose of specifically segmenting liver tumors. In our study, nnU-Net serves as a known well-performing baseline for us to compare against our lightweight custom architecture.

Recent submissions to the FLARE challenge also demonstrate the effectiveness of nnU-Net for cancer segmentation, specifically for abdominal and pan-cancer segmentation. The FLARE 2022 winning method used nnU-Net based methods to generate pseudo labels for unlabeled data and then trained a lightweight nnU-Net using labeled and pseudo-labeled data, maintaining segmentation accuracy while increasing computational efficiency (Huang et al., 2022). More recent methods, as of FLARE 2025, explore more lightweight options. Meng et al. designed a lightweight four-layer nnU-Net based framework with optimized patch size, spacing, and an adjusted training schedule for whole-body pan-cancer segmentation (Meng et al., 2025). They applied their model to whole-body pan-cancer segmentation. Our work differs by narrowing the scope to liver CT scans and liver tumor segmentation in particular.

Overall, our work expands on the established nnU-Net self-configuring baseline and is motivated by recent FLARE challenge methods which adapted nnU-Net for efficient cancer segmentation. Unlike these prior pan-cancer or pan-organ approaches, we focused specifically on supervised liver tumor segmentation in CT scans and utilized nnU-Net as the primary benchmark for evaluating our own lightweight custom model.

3 Method

3.1 Dataset

We used the Medical Segmentation Decathlon Liver Dataset (MSD Liver) for liver and tumor segmentation from contrast-enhanced CT scans. Each case contains a 3D CT volume with expert-annotated voxel-wise labels for liver and tumor regions. The dataset was preprocessed using the nnU-Net v2 preprocessing pipeline, which automatically performs resampling, intensity normalization, cropping, and patch generation. All experiments were evaluated using 5-fold cross-validation to obtain robust estimates of segmentation performance.

3.2 Baseline Model

Our baseline segmentation model was nnU-Net v2 using the 3D full-resolution configuration. In this setting, the network processes volumetric 3D image patches rather than individual 2D slices, allowing the model to capture spatial context across adjacent CT slices. The architecture follows a U-Net-style encoder-decoder structure with skip connections between corresponding resolution levels, During encoding, spatial resolution is progressively reduced while feature channels increase, enabling the network to learn hierarchical representations of anatomical structures. The decoder then reconstructs segmentation maps through upsampling and feature fusion with encoder outputs.

The full resolution configuration refers to training directly on high-resolution volumetric data without aggressive downsampling of the original CT images. Compared to 2D segmentation approaches, the 3D full resolution model an better utilze volumetric anatomical continuity and contextual information, which is essential for irregular or small tumor regions.

nnU-Net automatically configures optimal preprocessing parameters, patch size, batch size, data augmentation, network depth, and optimization settings based on the dataset features. For each fold in the 5-fold cross-validation setup, the model was trained on four folds and evaluated on the remaining held-out validation fold. Final evaluation metrics were computed from the complete valiation predictions generated after training, rather than from intermediate train metrics reported during optmization.

3.3 Comparison Model

To provide a lightweight comparison model, we additionally implemented a custom 2D U-Net model. Unlike the 3D nnU-Net model, this architecture processes individual axial CT slices independently rather than using full volumetric information. The model consists of a simplified encoder-decoder architecture with two downsampling stages, a bottleneck layer, and two upsampling stages connected through skip connections.

Each convolutional block contains two consecutive 3 x 3 convolutional layers followed by ReLU activations. Feature dimensions progressively increase from 32 to 128 channels in the encoder and decrease symmetrically in the decoder. Max-pooling layers were used for downsampling, while transposed convolutions were used for upsampling.

The lightweight 2D U-Net was trained using the same nnU-Net-preprocessed dataset and identical cross-validation split as the 3D nnU-Net experiments to ensure a fair comparison between methods. Validation performance was evaluated using the same held-out folds and segmentation metrics. This comparison model was designed to investigate the importance of volumetric 3D spatial context in liver tumor segmentation while also providing a substantially simpler and computationally lighter baseline architecture.

3.4 Evaluation Metrics

Segmentation performance was evaluated using class-wise Dice score for both liver and tumor segmentation. Dice score measures the overlap between predicted and ground-truth segmentation masks and is defined as:

$$Dice = \frac{2|P \cap G|}{|P| + |G|}$$

where P and G is the predicted and ground-truth voxel sets, respectively.

In addition to Dice score, the performance was further evaluated using Intersection over Union (IoU), sensitivity, and precision. These metrixs were computed using the number of true positives (TP), false positives (FP), and false negatives (FN). Sensitivity was computed as:

$$Sensitivity = \frac{TP}{TP + FN}$$

and precision was computed as:

$$Precision = \frac{TP}{TP + FP}$$

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These additional metrics provide insight into whether segmentation errors are primarily caused by missed tumors or false positive predictions. Final results are reported as the mean and standard deviation across all five cross-validation folds.

3.5 Qualitative and Volume-based Analysis

Qualitative segmentation analysis was performed using validation predictions aggregated across all five folds. Representative examples included well-segmented tumors, small or difficult tumor cases, and representative failure or false positive cases. For each example, CT images, ground truth masks, prediction masks, and overlay visualizations were generated.

To further analyze tumor segmentation behavior, we additionally created a tumor Dice vs. volume scatter plot using all validation cases across five folds. This analysis was used to investigate the relationship between lesion size and segmentation performance variability.

4 Experiment Results and Discussion

4.1 Results and Findings

The Dice similarity coefficient was used as the primary metric for segmentation accuracy, measuring the overlap between model segmentation and provided ground truths. Liver and tumor dice scores were determined from the results of 5-fold cross validation to assess each model's capacity to accurately segment and identify key areas of interest on CT scans. For the nnU-Net architecture, a mean Dice of 0.9569 ± 0.0066 for the liver and a mean Dice of 0.6446 ± 0.0719 for tumors were achieved (Table 1). These scores indicate a strong level of liver segmentation, but weaker tumor segmentation. This lower tumor Dice score is as expected due to the highly heterogeneous nature of tumors, and is in line with standard nnU-Net tumor segmentation performance. For the lightweight architecture, a mean Dice of 0.9206 ± 0.0866 for the liver and a mean Dice of 0.3672 ± 0.3063 for tumors were achieved. Liver segmentation accuracy is comparable across both models, but the lightweight model performs worse in tumor segmentation.

Table 1: nnU-Net vs Lightweight Model Fold-wise Liver and Tumor Segmentation Dice Scores ± SD

Fold	nnU-Net Liver	nnU-Net Tumor	Lightweight Liver	Lightweight Tumor
Fold 0	0.9654 ± 0.0122	0.7122 ± 0.2445	0.9409 ± 0.0126	0.3750 ± 0.3156
Fold 1	0.9623 ± 0.0223	0.5763 ± 0.3354	0.9316 ± 0.0463	0.3329 ± 0.3100
Fold 2	0.9506 ± 0.0526	0.6735 ± 0.2774	0.8858 ± 0.1690	0.4011 ± 0.3063
Fold 3	0.9544 ± 0.0425	0.7018 ± 0.2031	0.9370 ± 0.0312	0.4104 ± 0.2757
Fold 4	0.9518 ± 0.0409	0.5591 ± 0.3234	0.9069 ± 0.0608	0.3166 ± 0.3108
Overall Dice	**0.9569 ± 0.0066**	**0.6446 ± 0.0719**	**0.9206 ± 0.0866**	**0.3672 ± 0.3063**
Training Time	**369512 sec**		**6597 sec**	

IoU, sensitivity, and precision were used as secondary measures of segmentation accuracy in addition to Dice scores in determining fold-wise metrics (Table 2) and overall metrics (Table 3). Brief analysis reveals that across all 5 folds tested, the nnU-Net model outperformed our lightweight model across all secondary metrics. The higher sensitivity and precision values in particular indicate more optimal tumor segmentation, with more true positives, fewer false negatives, and fewer false positives. This suggests that the nnU-Net model identified tumors more consistently and had fewer instances of missed or incorrectly identified tumors. (Table 3).

Table 2: Tumor Detection and Segmentation Metrics: nnU-Net vs Lightweight Model (Fold-wise)

	nnU-Net			Lightweight Model		
Fold	**IoU**	**Sensitivity**	**Precision**	**IoU**	**Sensitivity**	**Precision**
0	0.5968	0.8351	0.8238	0.2805	0.5012	0.4291
1	0.4743	0.6339	0.7890	0.2450	0.4828	0.3695
2	0.5607	0.7877	0.7047	0.2979	0.5871	0.4348
3	0.5718	0.6590	0.8977	0.2976	0.3456	0.6942
4	0.4533	0.7761	0.9002	0.2327	0.3985	0.4536

Table 3: Tumor Detection and Segmentation Mean Metrics: nnU-Net vs Lightweight Model

Metric	nnU-Net	Lightweight Model
Tumor IoU	0.5314 ± 0.0635	0.2708 ± 0.2467
Sensitivity	0.7384 ± 0.0872	0.4633 ± 0.3313
Precision	0.8231 ± 0.0817	0.4759 ± 0.3733

A visual assessment of the tumor segmentations was conducted using both the lightweight model (Figure S1) and the nnU-Net model (Figure 1) to identify common failure modes. Figure 1 illustrates representative examples of optimal, poor, and failed segmentations, corresponding to Dice scores of 1.000, 0.399, and 0.000, respectively. Our analysis reveals that the modes primarily underperform in two scenarios: when the target tumor is small, or when the contrast between the tumor and the surrounding background is low. Small tumor regions are highly susceptible to misidentification, resulting in low Dice scores such as the poor segmentation example. Meanwhile, low-contrast can cause the model to miss the tumor region entirely, resulting in segmentation failure.

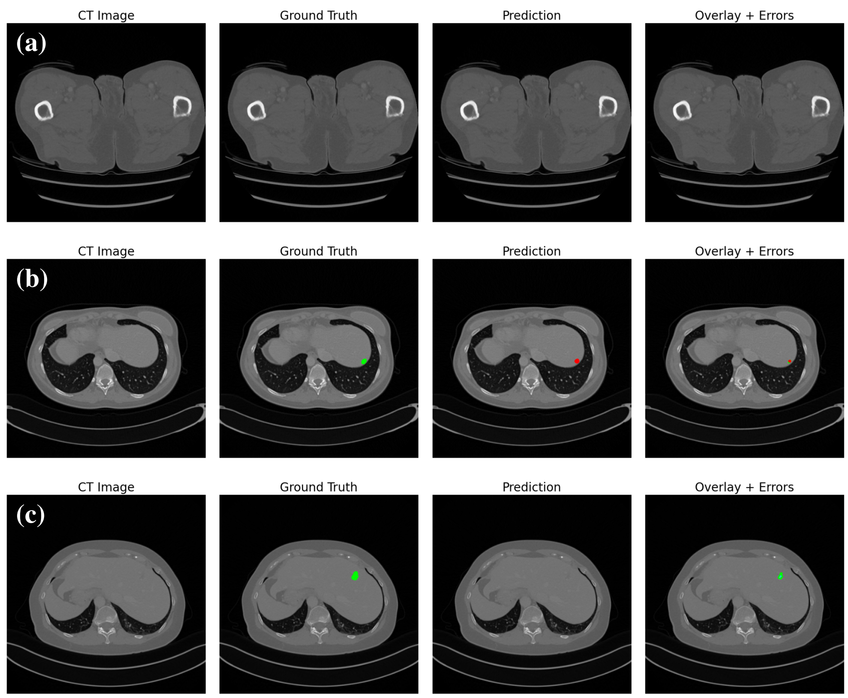
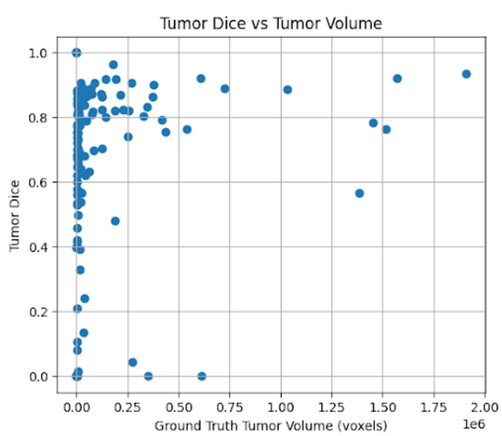
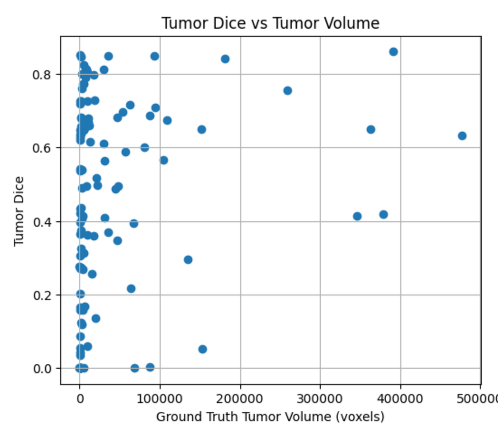


Figure 1: nnU-Net qualitative tumor segmentation results. Green contours indicate ground truth boundaries, red contours indicate predicted boundaries. a) good segmentation instance (Dice = 1.000). b) bad segmentation instance (Dice = 0.399). c) failed segmentation instance (Dice = 0.000)



(a) nnU-Net



(b) Lightweight Model

Figure 2: Tumor Dice score vs. tumor size comparison for the nnU-Net and lightweight models.

To support these qualitative assessments, tumor Dice scores were plotted against corresponding tumor volumes in voxels (Figure 2). The majority of the analyzed tumors fell within a smaller volumetric threshold between 0 and 0.25×10^6 voxels, which is also where the most variability in tumor Dice score is observed. As tumor volume increases, this variability also decreases, supporting the idea that small tumors are prone to misidentification. This pattern is observed in both the nnU-Net model, and the lightweight model.

4.2 Discussion

While the lightweight 2D U-net has a more efficient runtime, overall, nnU-Net had a stronger segmentation performance across both liver and tumor classes in comparison to the lightweight 2D U-Net. Both models have comparable liver Dice scores, but very contrasting tumor segmentation where nnU-Net achieved nearly double the mean Dice score of the 2D U-net. nnU-Net also outperformed the lightweight model in the secondary metrics of IoU, sensitivity, and precision, meaning that nnU-Net likely had fewer missed tumors and fewer false positives.

The difference in dice score is likely a result of several factors such as the nnU-Net's capability to process full volumetric patches which helps detect small or irregularly shaped tumors that could be ambiguous in a single 2D slice and therefore be missed by the lightweight 2D U-Net model. The lightweight model processes each slice independently and lacks this inter-slice context, hence why it has a higher rate of missed tumors and lower sensitivity.

The high variance in tumor Dice scores also suggests that the 2D model is less adaptable since it performs reasonably on larger or high-contrast tumors but struggles on smaller or more ambiguous cases. This is also shown in the scatter plot analysis, where there is high variability in lightweight Dice scores across tumor volumes.

Despite the weaker tumor segmentation, the lightweight model achieves comparable liver Dice scores to nnU-Net while training in significantly less time. The lightweight 2D U-Net is a good alternative when computational efficiency is prioritized and coarse liver segmentation is needed. For clinical applications requiring precise tumor delineation, such as surgical planning or treatment monitoring, it does not yet serve as a viable substitute for nnU-Net.

Appendix: Supplemental Figures

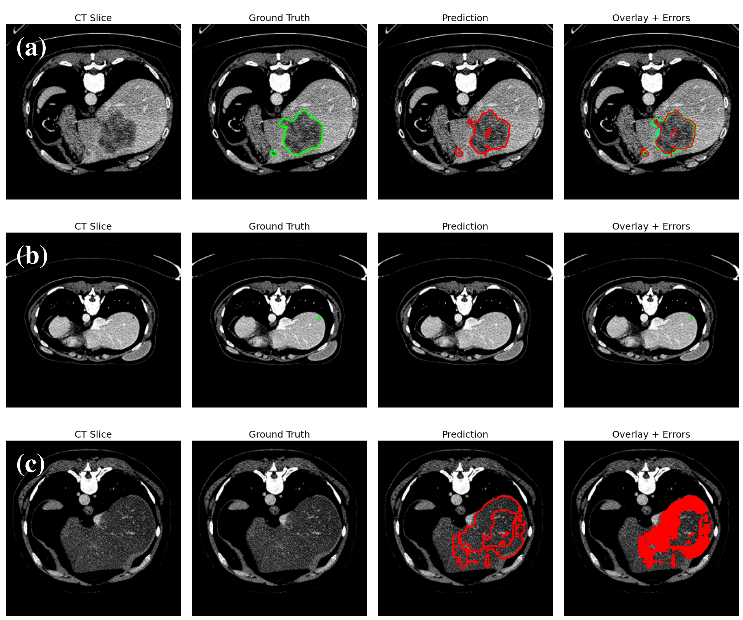


Figure S1: Lightweight qualitative tumor segmentation results. Green contours indicate ground truth boundaries, red contours indicate predicted boundaries. a) good segmentation instance (Dice = 0.861). b) false negative segmentation instance (Dice = 0.000). c) false positive segmentation instance (Dice = 0.000)

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